MACHINE LEARNING CASE STUDY | FINTECH & CREDIT RISK

Bank Loan Default Prediction Pipeline

How to build a leak-free Scikit-Learn Pipeline on 10,050 BDT loan records—and uncover what truly drives borrower default.

10,000	5 Models	0.7428	Top 5
Unique Loans Audited	5-Fold CV Benchmarked	Best Test ROC-AUC	Features Drive 85% Risk

Swipe to see the Data Audit, Pipeline Architecture, Benchmarks & Feature Insights --->

STEP 1: DATA AUDIT & LEAK-FREE PIPELINE

5 Hidden Traps Fixed Before Training

1. Target Leakage Removed (3 Columns)

days_to_default_resolution, delinquency_status & recovery_amount_bdt are only recorded AFTER a default. Keeping them causes fake 100% accuracy. Dropped alongside loan_id.

2. Duplicate Loan IDs Deduplicated (50 Rows)

10,050 raw rows contained 50 exact duplicates. Deduplicating prevents train-test contamination.

3. Sentinel Outliers Replaced with NaN

Replaced artificial error codes (age = 999 in 112 rows; credit_score = -1.0 in 94 rows) with NaN for median imputation.

4. Informative Missingness in Collateral Type (+0.031 AUC Gain)

4,012 missing collateral_type rows represent Unsecured ('None') loans with a 40.1% default rate (vs 20.4% for Property). Imputed as a dedicated 'None' category instead of mode imputation.

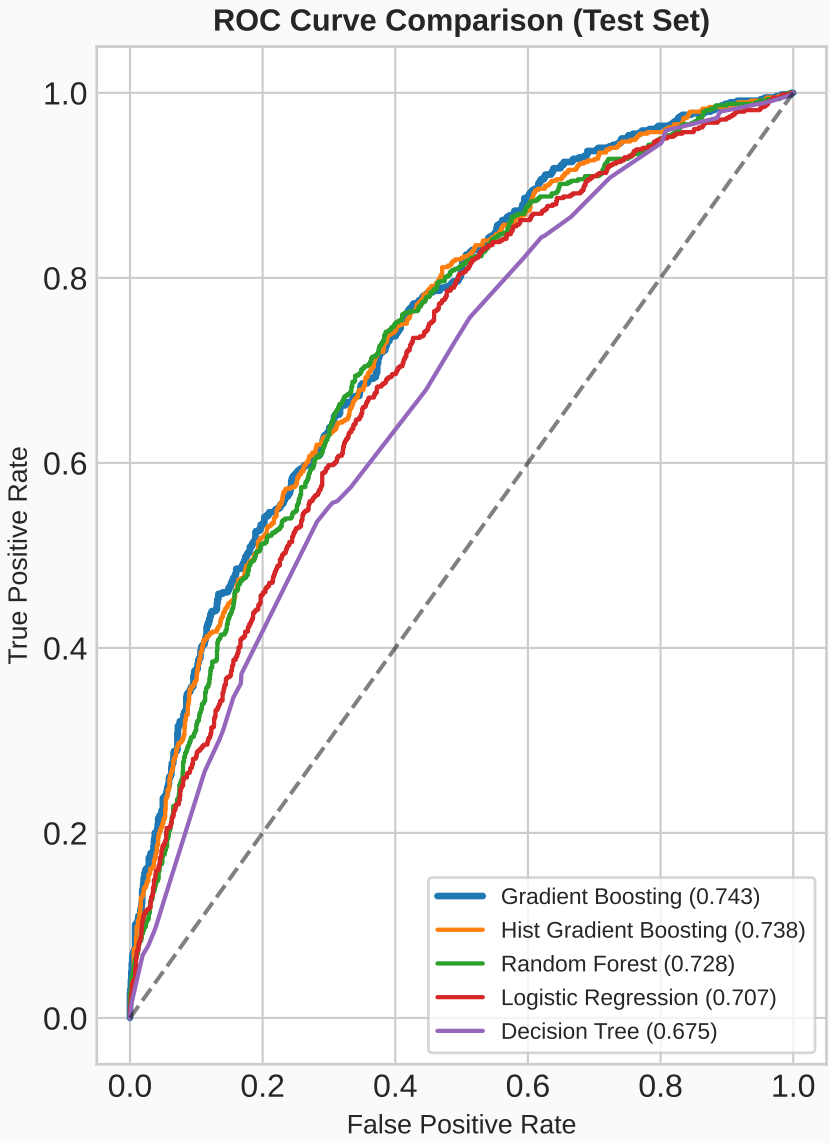
5. Unified Scikit-Learn ColumnTransformer Pipeline

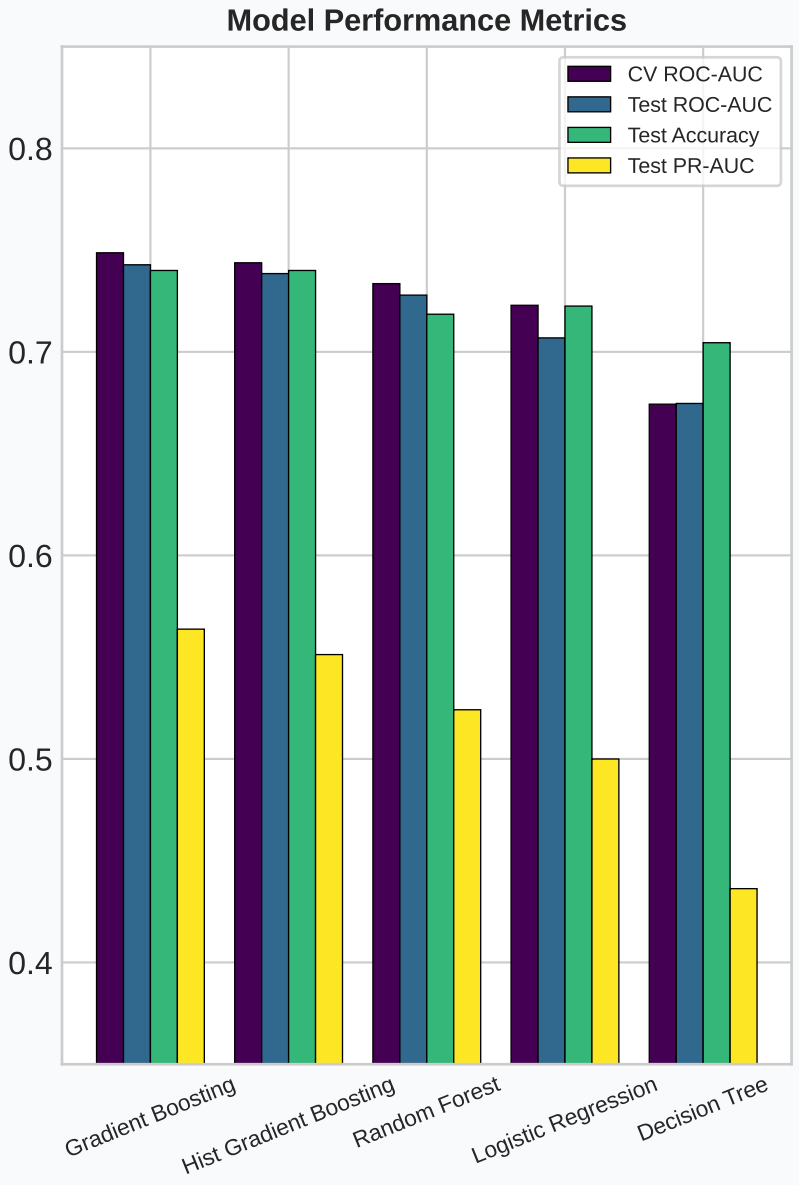
Numerical (13 cols): Median Imputer -> StandardScaler | Categorical (4 cols): Mode/Constant Imputer -> OneHotEncoder.

STEP 2: MULTI-MODEL BENCHMARKING (5-FOLD STRATIFIED CV)

Gradient Boosting Wins Across All Key Metrics

- Best Test ROC-AUC: 0.7428 | 5-Fold CV ROC-AUC: 0.7487 (±0.0064)
- Best Test PR-AUC: 0.5638 | Test Accuracy: 74.00% | Test F1-Score: 0.3765 (0.5363 @ Threshold=0.32)

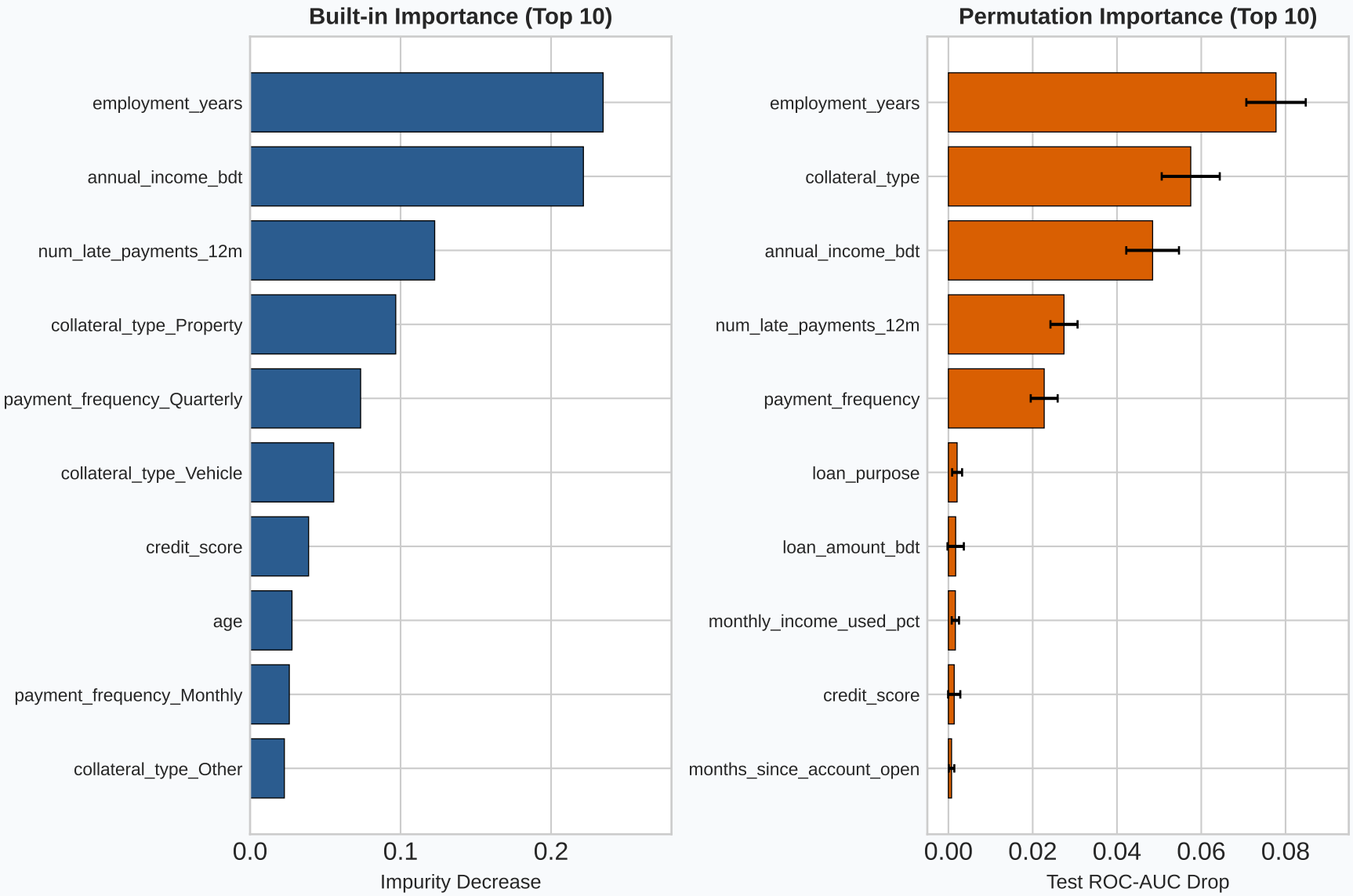




STEP 3: ATTRIBUTE IMPORTANCE & BANKING IMPACT

5 Attributes Drive >85% of Default Predictions

1. Employment Years (+0.0777 AUC drop) 2. Collateral Type (+0.0575) 3. Annual Income (+0.0485)
4. Late Payments 12m (+0.0274) 5. Payment Frequency (+0.0227) | Gender = 0.0000 (Fair & Unbiased)



Full Python & Scikit-Learn Pipeline code available in the GitHub repository link in the comments!